Міністерство освіти і науки України
Національний технічний університет України
«Київський політехнічний інститут імені Ігоря Сікорського»
Факультет інформатики та обчислювальної техніки
Кафедра обчислювальної техніки

Лабораторна робота 4

з дисципліни
«Бази даних»

на тему

« **Аналітичні SQL-запити(OLAP)**»

Виконали:

студенти групи ІМ-55

Бабенко Ярослав Володимирович
Резніков Нікіта Сергійович

Номер в списку - 3, 24

Перевірив:
Хмельницький А. А.

SQL-скрипти:

```
SELECT
   COUNT(*) AS total_reviews,
   SUM(rating) AS sum_ratings,
   AVG(rating) AS avg_rating,
   MIN(rating) AS min_rating,
   MAX(rating) AS max_rating
FROM review;

SELECT
   poem_status,
   COUNT(*) AS poems_count
FROM poem
GROUP BY poem_status;

SELECT
   poem_id,
   COUNT(*) AS reviews_count,
   SUM(rating) AS total_points
FROM review
GROUP BY poem_id
ORDER BY total_points DESC;

SELECT
   poem_id,
```

```
    COUNT(*) AS reviews_count
FROM review
GROUP BY poem_id
HAVING COUNT(*) > 1;

SELECT
    p.title,
    p.poem_status,
    per.username
FROM poem p
INNER JOIN person per ON p.owner_id = per.person_id;

SELECT
    p.title,
    r.rating,
    r.comment
FROM poem p
LEFT JOIN review r ON p.poem_id = r.poem_id;

SELECT
    p.title,
    COUNT(r.poem_id) AS reviews_count
FROM poem p
LEFT JOIN review r ON p.poem_id = r.poem_id
GROUP BY p.poem_id, p.title
```

```
ORDER BY reviews_count DESC;


SELECT
  per.username,
  (SELECT COUNT(*) FROM poem p WHERE p.owner_id = per.person_id)
AS poems_count
FROM person per;


SELECT
  poem_id,
  title
FROM poem
WHERE poem_id IN (
  SELECT poem_id
  FROM review
  WHERE rating = 5
);


SELECT
  poem_id,
  AVG(rating) AS poem_avg
FROM review
GROUP BY poem_id
HAVING AVG(rating) > (
  SELECT AVG(rating)
```

```
  FROM review
);



SELECT
  COUNT(*) AS total_poems,
  SUM(poem_id) AS sum_ids,
  AVG(poem_id) AS avg_id,
  MIN(published_at) AS first_published,
  MAX(published_at) AS last_published
FROM poem;

SELECT
  visibility_status,
  COUNT(*) AS collections_count
FROM collection
GROUP BY visibility_status;

SELECT
  collection_id,
  COUNT(*) AS poems_count,
  MAX(added_at) AS last_added
FROM collection_poem
GROUP BY collection_id
```

```
ORDER BY poems_count DESC;

SELECT
   collection_id,
   COUNT(*) AS poems_count
FROM collection_poem
GROUP BY collection_id
HAVING COUNT(*) > 1;

SELECT
   c.title,
   c.visibility_status,
   per.username
FROM collection c
INNER JOIN person per ON c.person_id = per.person_id;

SELECT
   per.username,
   pa.poem_id,
   pa.contribution_type
FROM poem_author pa
RIGHT JOIN person per ON pa.user_id = per.person_id;

SELECT
   c.title,
```

```
    COUNT(cp.poem_id) AS poems_count
FROM collection c
FULL JOIN collection_poem cp ON c.collection_id = cp.collection_id
GROUP BY c.collection_id, c.title
ORDER BY poems_count DESC;

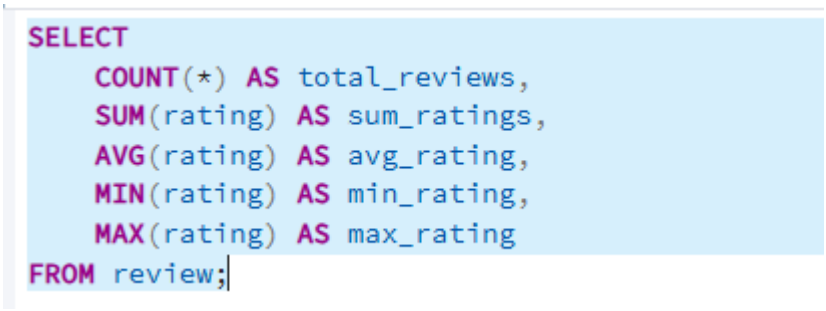
SELECT
    c.title,
    (SELECT COUNT(*) FROM collection_poem cp WHERE cp.collection_id = c.collection_id) AS poems_count
FROM collection c;

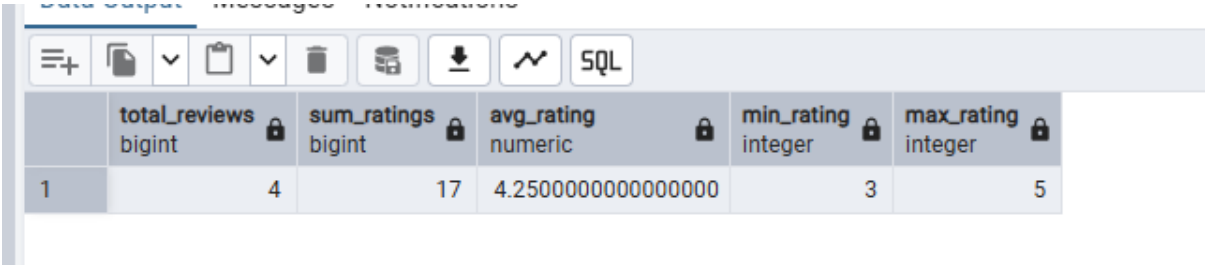
SELECT
    poem_id,
    title
FROM poem
WHERE poem_id IN (
    SELECT poem_id
    FROM collection_poem
    WHERE collection_id = 1
);

SELECT
    collection_id,
    COUNT(*) AS poems_count
```

```
FROM collection_poem
GROUP BY collection_id
HAVING COUNT(*) > (
   SELECT COUNT(*)
   FROM collection_poem
   WHERE collection_id = 2
);
```

Скріншоти:





	total_reviews bigint	sum_ratings bigint	avg_rating numeric	min_rating integer	max_rating integer
1	4	17	4.2500000000000000	3	5

```sql
SELECT
    COUNT(*) AS total_reviews,
    SUM(rating) AS sum_ratings,
    AVG(rating) AS avg_rating,
    MIN(rating) AS min_rating,
    MAX(rating) AS max_rating
FROM review;

SELECT
    poem_status,
    COUNT(*) AS poems_count
FROM poem
GROUP BY poem_status;

SELECT
    poem_id,
    COUNT(*) AS reviews_count,
    SUM(rating) AS total_points
FROM review
GROUP BY poem_id
ORDER BY total_points DESC;

SELECT
    poem_id,
    COUNT(*) AS reviews_count
FROM review
GROUP BY poem_id
HAVING COUNT(*) > 1;

SELECT
    p.title,
    p.poem_status,
    per.username
FROM poem p
INNER JOIN person per ON p.owner_id = per.person_id;

SELECT
    p.title,
    r.rating,
    r.comment
FROM poem p
LEFT JOIN review r ON p.poem_id = r.poem_id;

SELECT
    p.title,
    COUNT(r.poem_id) AS reviews_count
```

```
32      p.poem_status,
33      per.username
```

Data Output Messages Notifications

	collection_id integer	poems_count bigint
1	1	2